

Calc 1 Lecture 21: Optimization

We've seen how to find max and min values of functions.
Now we explore the applications of this to "real-world" problems.

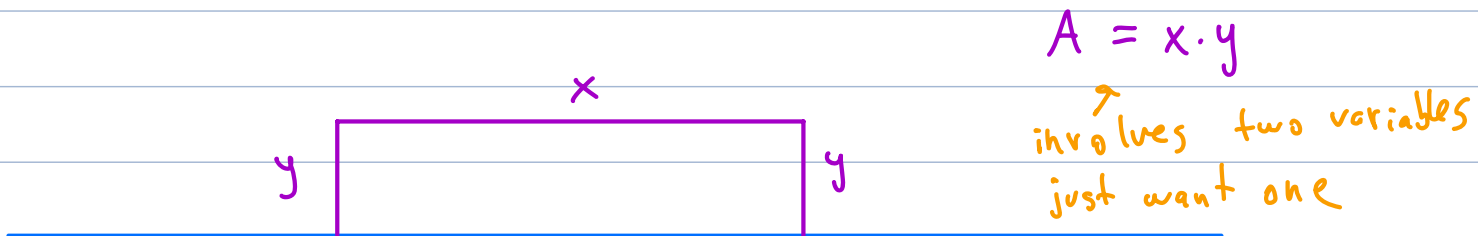
constraint

Example A farmer has 2400 ft of fencing and wants to fence off a rectangular field that borders a straight river. He needs no fence along the river.

What are the dimensions of the field that has the largest area?

Idea: express this as finding max of some function

Step 0: draw picture. Identify relevant variable(s) and function we want to optimize



Step 1: To make it a function of one variable, use given constraint to eliminate a variable.

$$2400 = x + 2y \Rightarrow x = 2400 - 2y$$

$$A = x \cdot y = (2400 - 2y) \cdot y = 2400y - 2y^2 = A(y)$$

Step 2: optimize function by finding critical points.

$$A(y) = 2400y - 2y^2$$

$$A'(y) = 2400 - 4y$$

$$A'(y) = 0 \Rightarrow 2400 - 4y = 0 \Rightarrow 4y = 2400 \Rightarrow y = 600$$

$$x = 2400 - 2y \Rightarrow x = 1200$$

Step 3: justify why your critical point is the optimum you are looking for (in our case, max)

Two approaches:

Approach 1: if our variables are defined over a closed interval, can use extreme value Theorem

$\Rightarrow$ absolute max/min is either at critical point or boundary point.

in our case, no negative lengths

$$\begin{aligned} x \geq 0 \} & x = 2400 - 2y \geq 0 \Rightarrow 2y \leq 2400 \Rightarrow y \leq 1200 \\ y \geq 0 \end{aligned}$$

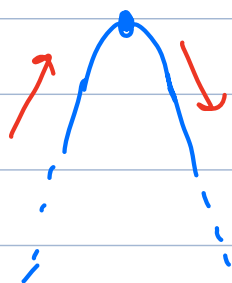
$$\Rightarrow 0 \leq y \leq 1200 \quad \text{closed interval}$$

y	$A(y) = 2400y - 2y^2 = (2400 - 2y) \cdot y$
0	0
600	$1200 \cdot 600 > 0$

absolute max of A over $[0, 1200]$

120g | 0

Approach 2: if we have exactly one critical point
then enough to find local max/min] either 1st deriv. test or
it will automatically be a global max/min. 2nd deriv. test



derivative cannot change
sign at any other point.

Second deriv. test: $A'(y) = 2400 - 4y$

$$A''(y) = -4 < 0$$

by 2nd deriv. test $\Rightarrow$ $y = 600$ is
a local max

$\Rightarrow$ $y = 600$ is a
global max
because it's the only
critical point

area 1000 cm^3

Example A cylindrical can is to be made to hold 1 L of oil.

Find the dimensions that will minimize the cost of the metal to manufacture the can.
proportional to surface area

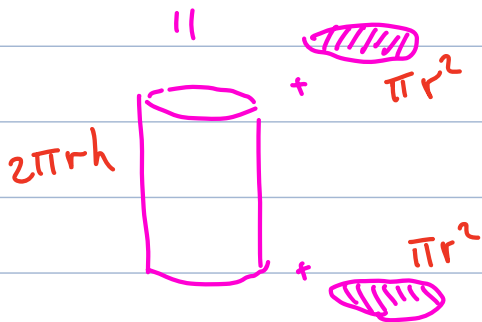
Step 0: set up. sketch picture, define relevant variables and functions

state constraint



minimize surface area

$$SA = 2\pi r \cdot h + 2\pi r^2$$



constraint: $V = 1 \text{ L} = 1000 \text{ cm}^3$

$$\underbrace{\pi r^2}_{\text{base}} \cdot \underbrace{h}_{\text{height}} = 1000$$

Step 1: use constraint to rewrite objective function in terms of one variable
function want to optimize

constraint: $\pi r^2 h = 1000$

$$h = \frac{1000}{\pi r^2}$$

$$SA = 2\pi r \cdot h + 2\pi r^2 = \frac{2\pi r \cdot 1000}{\pi r^2} + 2\pi r^2 = f(r)$$

$$\frac{2000}{r} + 2\pi r^2 = f(r)$$

Step 2: Optimize by finding critical points

$$f(r) = 2000 \cdot r^{-1} + 2\pi r^2 \Rightarrow f'(r) = \underbrace{-2000 \cdot r^{-2} + 4\pi r}_{\cdot r^2 \downarrow} = 0$$
$$-2000 + 4\pi r^3 = 0$$

$$4\pi r^3 = 2000$$

$$r^3 = \frac{500}{\pi}$$

critical
point

$$r = \sqrt[3]{\frac{500}{\pi}}$$

$$h = \frac{1000}{\pi r^2} = \frac{1000}{\pi \left(\frac{500}{\pi}\right)^{2/3}}$$

Step 3: Justify critical pt is absolute min

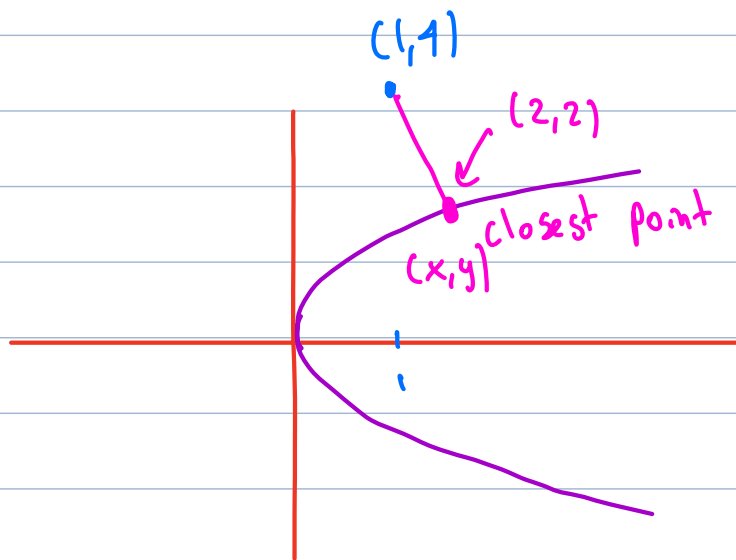
$$f''(r) = \frac{d}{dr} (-2000 \cdot r^{-2} + 4\pi r) = 4000 r^{-3} + 4\pi > 0$$

when $r > 0$

$\Rightarrow$ critical point is local min by 2nd deriv. test

$\Rightarrow$ so it is global min because it's the only critical point

Example Find the point on the parabola $y^2 = 2x$ that is closest to the point $(1, 4)$.



minimize distance between $(1, 4)$ and (x, y)

$$d = \sqrt{(x-1)^2 + (y-4)^2} \quad (\text{distance formula})$$

constraint: $y^2 = 2x$ $x = \frac{y^2}{2}$

$$d = \sqrt{\left(\frac{y^2}{2} - 1\right)^2 + (y-4)^2}$$

$$d^2 = \left(\frac{y^2}{2} - 1\right)^2 + (y-4)^2 = f(y)$$

minimizing d
equivalent to
minimizing d^2

minimize $f(y)$

Find critical points:

$$f'(y) = 2\left(\frac{y^2}{2} - 1\right) \cdot \underbrace{\frac{d}{dy}\left(\frac{y^2}{2} - 1\right)}_{\frac{1}{2} \cdot 2y = y} + 2(y-4) \cdot \underbrace{\frac{d}{dy}(y-4)}_1$$

$$= 2\left(\frac{y^2}{2} - 1\right) \cdot y + 2(y-4)$$

$$(y^2 - 2)y + 2y - 8 = y^3 - \cancel{2y} + \cancel{2y} - 8 = \overbrace{y^3 - 8}^{f'(y)}$$

critical points: $y^3 - 8 = 0 \Rightarrow y^3 = 8 \Rightarrow \underline{y = 2}$

$$x = \frac{y^2}{2} \Rightarrow x = \frac{2^2}{2} = 2$$

Final answer: $\boxed{(x, y) = (2, 2)}$

Justify why critical point is a min:

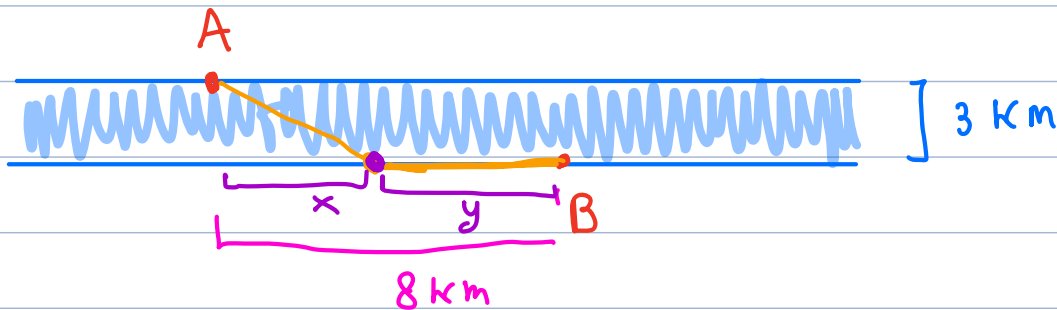
$$f'(y) = y^3 - 8 \Rightarrow f''(y) = 3y^2 > 0$$

$\Rightarrow y = 2$ is a local min $\Rightarrow$ global min because it's the only critical point

Example A man launches his boat from point A on a bank of a straight river 3 km wide and wants to reach point B , 8 km downstream on the opposite bank. If he can row 6 km/h and run 8 km/h where shall he land to reach B as soon as possible?
(assume speed of water is negligible)

minimize time

Step 0: set up

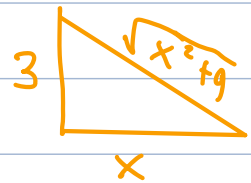


$$\text{speed} = \frac{\text{distance}}{\text{time}}$$

$$\text{time} = \frac{\text{distance}}{\text{speed}}$$

$$T = T_{\text{row}} + T_{\text{run}}$$

$$T = \frac{\sqrt{x^2 + 9}}{6} + \frac{y}{8}$$



Constraint: $x + y = 8 \Rightarrow y = 8 - x$

$$T = \frac{\sqrt{x^2 + 9}}{6} + \frac{8 - x}{8} = f(x)$$

optimize! $f'(x) = \frac{1}{6} \underbrace{\frac{d}{dx} \sqrt{x^2 + 9}}_{\frac{1}{\sqrt{x^2 + 9}} \cdot 2x} + \frac{1}{8} \underbrace{\frac{d}{dx} (8 - x)}_{-1}$

$$f'(x) = \frac{x}{6\sqrt{x^2+9}} - \frac{1}{8} = 0$$

$$\frac{x}{6\sqrt{x^2+9}} = \frac{1}{8}$$

cross multiply

$$8x = 6\sqrt{x^2+9}$$

square

$$64x^2 = 36 \cdot (x^2+9)$$

$$64x^2 = 36x^2 + 36 \cdot 9$$

$$28x^2 = 36 \cdot 9$$

$$x^2 = \frac{36 \cdot 9}{28} = \frac{9 \cdot 9}{7} = \frac{9^2}{7}$$

$$x = \frac{9}{\sqrt{7}} \approx 3.4 \text{ km}$$

Answer: land ≈ 3.4 km downstream

Justifying optimum:

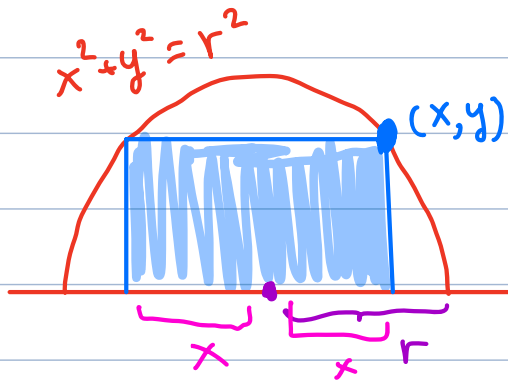
$0 \leq x \leq 8$ (over a closed interval)

$\Rightarrow$ can use EVT.

x	T
0	≈ 1.5
$9/\sqrt{7}$	≈ 1.3 min
8	≈ 1.4

Example Find the area of the largest rectangle that can be inscribed in a semicircle of radius r maximize area

step 0:



relevant variables
objective function
constraint

$$\left. \begin{array}{l} x, y \\ A = 2xy \\ x^2 + y^2 = r^2 \end{array} \right\}$$

$$A = \underbrace{\text{width}}_{2x} \cdot \underbrace{\text{height}}_y = 2xy$$

step 1: use constraint to rewrite objective function in terms of only one variable.

$$x^2 + y^2 = r^2 \quad \left. \vphantom{x^2 + y^2 = r^2} \right\} \text{constraint}$$

$$y^2 = r^2 - x^2 \Rightarrow y = \sqrt{r^2 - x^2}$$

$$A = 2xy = 2x \sqrt{r^2 - x^2}$$

step 2: optimize. maximize $A \Leftrightarrow$ maximize A^2

$$A^2 = (2x \sqrt{r^2 - x^2})^2 = 4x^2 (\sqrt{r^2 - x^2})^2 = 4x^2 (r^2 - x^2)$$

$$= 4x^2 r^2 - 4x^4 = f(x)$$

$$f'(x) = 8xr^2 - 16x^3 = 0$$

$$x(8r^2 - 16x^2) = 0 \Rightarrow x=0 \leftarrow \text{not relevant area 0.}$$

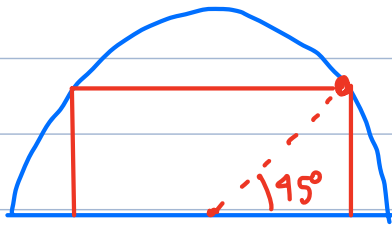
$$\text{or } \underline{8r^2 - 16x^2 = 0}$$

$$8r^2 = 16x^2$$

$$\boxed{x = \frac{r}{\sqrt{2}}}$$

$$\Leftrightarrow \left(\frac{r}{2} \right)^2 = x^2$$

$$y = \sqrt{r^2 - x^2} = \sqrt{r^2 - \frac{r^2}{2}} = \sqrt{\frac{r^2}{2}} = \boxed{\frac{r}{\sqrt{2}} = y}$$



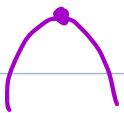
$$A = 2xy = 2 \frac{r}{\sqrt{2}} \frac{r}{\sqrt{2}} = 2 \cdot \frac{r^2}{2} = \boxed{r^2}$$

Justification that $\boxed{x = \frac{r}{\sqrt{2}}}$ is a max:

$$f'(x) = 8xr^2 - 16x^3 \Rightarrow f''(x) = 8r^2 - 48x^2$$

$$f''\left(\frac{r}{\sqrt{2}}\right) = 8r^2 - \frac{48r^2}{2}$$

$$= 8r^2 - 24r^2 = -16r^2 < 0$$



$\Rightarrow \frac{r}{\sqrt{2}}$ is a local max

$\Rightarrow$ global max because it's the only critical point